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Sub:ATD

Subject code: BTMC506

Class: B.Tech., Sem-V

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Q. No.	Question
UNIT NO.1	
1.	What is meant by term fuel? Classify fuels in detail.
2.	Discuss the construction and working of Bombs calorimeter with neat sketch and thus derive the formula of HCV in Bomb Calorimeter
3.	Calculate the Gross and Net calorific values of a coal having the following compositions, C = 80 %, H ₂ = 8 %, O ₂ = 8 %, S = 2 % and ash = 2%. Latent heat of steam is = 2465 kJ/kg.
4.	Explain briefly, the method used to determine the higher calorific value of the liquid
5.	Determination of minimum air required for complete combustion of 1 kg of solid or liquid fuels
6.	How is flue gas analyses using Orsat's apparatus?
7.	The ultimate analysis by percentage of weight of a sample of coal is as follows C=65, H=5, O= 20, S=2, Find the calorific value of the coal by Dulong's Formula
8.	A single cylinder was supplied with a gas having the following percentage volumetric analysis; CO = 5, CO ₂ = 10, H ₂ = 50, CH ₄ = 25, N ₂ = 10. The percentage volumetric analysis of dry gases was CO ₂ = 8, O ₂ = 6 and N ₂ = 86. Determine the air-fuel ratio by volume.
9.	A sample of fuel has the following percentage composition : Carbon = 86 per cent ; Hydrogen = 8 per cent ; Sulphur = 3 per cent ; Oxygen = 2 per cent ; Ash = 1 per cent. For an air-fuel ratio of 12 : 1, calculate : (i) Mixture strength as a percentage rich or weak. (ii) Volumetric analysis of the dry products of combustion.
10.	A coal sample gave the following analysis by weight, Carbon 85 per cent, Hydrogen 6 per cent, Oxygen 6 per cent, the remainder being incombustible. Determine minimum weight of air required per kg of coal for chemically correct composition.

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11.	The following data obtained during refer to an experimental determination of the calorific value of a sample of fuel containing 88%C, and 4.5% H ₂ , Weight of coal = 2.1 gm, weight of fuse wire 0.025 gm, of calorific value 5300 J/gm, Weight of water in the calorimeter = 1650gm, water equivalent of calorimeter =380gm, observed temperature rise =2.98 °C, Cooling correction = 0.031 °C. Find the higher & Lower calorific value of fuel
12.	The following particulars refer to an experimental determination of the calorific value of a sample of coal containing 88% C and 4.2 % H ₂ ,Weight of coal =0.848gm, weight of fuse wire 0.027gm,of calorific value 6700 J/gm, weight of water in the calorimeter = 1950gm,water equivalent of calorimeter =380gm, observed temperature rise = 3.06 °c, cooling correction = + 0.017°c. Find the higher & lower calorific values of the coal.
13.	Define 1. Calorific value of fuels 2. HCV 3. LCV 4. Theoretical air & 5. Excess air & 6. Stoichiometric air fuel (A/F) Ratio
14.	Explain construction & working of bomb calorimeter
15.	Calculate the Gross and Net calorific values of a solid fuel having 80% of carbon & 20% of hydrogen. Latent heat of steam is = 587 cal/gm.
UNIT NO.2	
1.	Explain following boiler mountings with neat sketch a) Water level indicator b) Dead weight safety valve c) Feed check valve d) Fusible plug e) Pressure gauge
2.	Sketch and describe a Cochran boiler. What are its special features
3.	Sketch and describe Bensons boiler. What are its special features
4.	What is the difference between FTB & WTB?
5.	Which factors you will consider while selecting a steam boiler?
6.	Define steam boiler and how boilers are classifies?
7.	Described IBR Boilers in details

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8.	Explain different accessories used in steam boilers with neat sketch
9.	What are the different sources of heat loss in a boiler plant?
10.	Explain following terms with their significance Evaporation, equipment evaporation, boiler efficiency, boiler trial and heat balance
11.	Explain procedure to draw heat balance sheet for a boiler plant
12.	Compare nature draught and artificial draught
13.	Differentiate between force and induced draught
14.	What is draught? How to Classification of draught
15.	Explain efficiency of the chimney and draught losses
16.	Sketch and describe following boilers 1. Babcock or Wilcox Boiler 2. Lancashire Boiler 3. Locomotive Boiler 4. Lamont Boiler
17.	A boiler evaporates 3.6 kg of water per kg of into dry saturated steam at 10 bar. Find the equivalent evaporation if feed water temperature is 32 °C
18.	The following readings were obtained during a boiler trial of 6 hour duration. Mean Steam Pressure = 12 bar, Mass of steam generated = 40000Kg, Mean dryness fraction = 0.85 Mean feed water temp = 30° C Coal Used = 4000kg Calorific value of coal = 33400 kJ/kg Calculate: (1) Factor of equivalent evaporation(F_e) (2) Equivalent evaporation (M_e) (3) Efficiency of boiler

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19.	The boiler trial following observations are recorded mass of fuel is 1520 kg/hr. The temperature of feed water is 30 deg. C. Dryness fraction of steam is 0.95. The pressure of steam is 8.5 bar, Coal burns per hour = 200 kg, CV of coal is 27300 kJ/kg. The un burnt coal collected is 60 kg/hr with a CV of 2000 kJ/kg. The mass of flue gases is 17.73 kg/kg of coal burnt. The temperature of flue gases is 330 deg. C. The boiler room temperature is 27 deg. C. Specific heat of the flue gases 1 kJ/kg K. Draw Boiler heat balance sheet per kg of fuel burnt and calculate the efficiency of the boiler.
20.	Q. In a Boiler trial, the following observations were made-pressure of the steam 12 bar, Rate of steam evaporation is 620kg/hr, Fuel used is 65kg/hr, moisture in fuel is 2% by mass, mass of dry glue gases is 10kg/kg, of fuel, Lower C.V. is 3000kj/kg, Temp. of flue gases is 320c, Temp. of boiler house is 26c, Feed water temp. is 60c, cp is 1.005kj/kg k, Dryness fracyion is 0.96, Draw heat balance sheet.
21.	The following readings were obtained during a boiler trial of 1 hour duration. Mean Steam Pressure = 7.6 bar, Mass of steam generated = 5000Kg, Mean dryness fraction = 0.98 Mean feed water temp = 51° C Coal Used = 650kg Calorific value of coal = 30500 kJ/kg Calculate: (1) Factor of equivalent evaporation (Fe) (2) Equivalent evaporation (Me) (3) Efficiency of boiler
22.	The following readings were obtained during a boiler trial of 6-hour duration. Mean Steam Pressure = 12 bar, Mass of steam generated = 40000kg, Mean dryness fraction = 0.9, Mean feed water temp = 31 °C, Coal Used = 4000kg, Calorific value of coal = 33450 kJ/kg Calculate: (1) Factor of equivalent evaporation (2) Equivalent evaporation (3) Efficiency of boiler

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UNIT NO. 3	
1.	Described the different operation of Rankine cycle. Derive also the expression for its efficiency
2.	Explain the various operation of a Carnot cycle. Also represent it on a T-S and P-V diagrams.
3.	Explain Rankine cycle with Reheat and Regeneration
4.	Explain Stirling cycle with T-S and P-V diagrams.
5.	Explain Brayton Cycle with T-S and P-V diagrams. Also calculate Air- Standard Efficiency
6.	Explain Closed & Open Cycle Gas Turbine
7.	Explain Brayton cycle with Reheat and Regeneration
8.	Dry saturated steam at 17.5 bar enters the turbine of a steam power plant and expands to the condenser pressure of 0.75 bar. Determine the Carnot and Rankine cycle efficiencies. Also find the work ratio of the Rankine cycle
9.	In a Rankine cycle, the steam of inlet to turbine is saturated at a pressure of 35 bar and the exhaust pressure is 0.2 bar. Determine (a) The pump work, (b) The turbine work, (c) The Rankine efficiency, (d) The condenser heat flow, and (e) The dryness at the end of the expansion. Assume flow rate of 9.5 kg/sec
10.	A Steam power plant has steam at a Pressure of 40 bar and temperature 400° C and exhausted in to a condenser where, a pressure of 0.05 bar is maintained. The mass flow rate of steam is 160 kg/sec. Determine : • The Rankine cycle efficiency • Power developed • specific steam consumption • The heat rejected into the condenser per hour • Carnot efficiency

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11.	In a single heater regenerative cycle the steam enters the turbine at 30 bar , 400 °C and the exhaust pressure is 0.10 bar. The feedwater heater is a direct contact type which operates at 5 bar . find (a) the efficiency and the steam rate of the cycle; (b) the increase in mean temperature of heat addition, efficiency and the steam rate, as compared to the Rankine cycle (without regeneration). neglect pump work.
12.	• In a Brayton cycle based power plant, the air at the inlet is at 27°C, 0.1 MPa. The pressure ratio is 6.25 and the maximum temperature is 800°C. Find (a) the compressor work per kg of air (b) the turbine work per kg of air (c) the heat supplied per kg of air, and (d) the cycle efficiency.
13.	Air enters the compressor of a gas turbine plant operating on Brayton cycle at 1 bar, 27°C. The pressure ratio in the cycle is 6. If $W_t = 2.5 W_c$ where W_t and W_c are the turbine and compressor work respectively, calculate the maximum temperature and the cycle efficiency.
14.	In air standard Brayton cycle, the enters the compressor at 1 bar and 15 °C. the pressure leaving the compressor is 5 bar and the maximum temperature in the cycle is 900 °C. find the following: (i) compressor and expander work per kg of air (b) the cycle efficiency .
15.	In a gas turbine power plant working on Brayton cycle, the inlet air temperature is 30 °C and pressure is 1.0 bar, the pressure ratio is 6.25 and the maximum temperature is 827 °C. Find (a) the compression work (b) the turbine work, (c) the cycle efficiency and (d) the work ratio Compared the efficiency with Carnot cycle operating between the same temperature limits.
16.	What do you mean by a supersaturated flow? Explain with the help of h-s diagram
17.	Derive the equation for condition of maximum discharge through a steam nozzle.
18.	Define steam nozzle and explain in details types of nozzles
19.	With h-s graph explain effect of friction between nozzle surface and steam
20.	Explain general relationship between area, velocity and pressure of steam nozzle
21.	Explain Flow of steam through nozzles

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22.	Dry saturated steam at a pressure of 8.5 bar enters a convergent-divergent [nozzle, and leaves at a pressure of 15 bar. ; If the flow is frictionless adiabatic and the corresponding expansion index is 1.135, \ find using steam tables, the ratio of the cross-sectional area at exit to that at the throat,
23.	A steam nozzle is to be designed for a mass flow rate of 8 kg/s of steam from 8 bar and 350°C to a back pressure of 1 bar. The nozzle efficiency is 0.93 and the frictional loss is assumed to take place in the diverging portion of the nozzle only. Determine the throat and exit area.
24.	A Nozzle expands steam from 12 bar and 250°C to 6 bar. Is the Nozzle convergent or convergent - divergent (C-D) ? Neglecting the initial velocity. Find the minimum area of the nozzle to flow 2 kg/s of steam under given conditions. Assume the expansion of steam is isentropic. Calculate the actual throat area if the co-efficient of discharge is 0.98.
25.	Dry saturated steam enters a nozzle at pressure of 10 bar and with an initial velocity of 90 m/sec. The outlet pressure is 6 bar and outlet velocity is 435 m/sec. The heat loss from the nozzle is 63 kJ per kg of steam flow. Calculate the dryness fraction and the area at the exit, if the area at the inlet is 1256 cm ² .
26.	Dry saturated steam at a pressure of 10 bar enters a C-D Nozzle and leaves at a pressure of 1.8 bar. If the flow is adiabatic and frictionless, determine i) The exit velocity of steam iii) Ratio of cross-section at exit and that at throat Assume the index of adiabatic expansion to be 1.133.
27.	Dry saturated steam at a pressure of 15 bar enters in a nozzle and is discharged at a pressure – of 1.5 bar. Find the final velocity of steam, when the initial velocity of steam is negligible. If 10% of the heat drop is lost in friction, find the percentage reduction in the final velocity. --- (6)
UNIT NO.4	
1.	Define steam turbine and explain the classification of steam turbines
2.	Describe the construction and working of Impulse steam turbine
3.	Explain velocity compounding of impulse turbine with neat sketch
4.	What is Compounding of steam turbines? Explain pressure compounding steam turbine

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5.	What are the types of compounding ? Explain Pressure-velocity compounding of impulse turbine with neat sketch
6.	Explain velocity diagram for impulse steam turbine with combined velocity diagram
7.	Differentiate between the impulse and reaction steam turbine
8.	Explain advantages of steam turbine
9.	Describe the construction and working of reaction steam turbine
10.	Explain Losses in steam turbines
11.	Define following terms 1.Blade/Diagram efficiency 2.Gross stage efficiency 3.Degree of reaction
12.	A steam impulse turbine has a mean blade speed of 200 m/s. The nozzles are inclined at 20° to the plane of rotation of the blades. The steam velocity from nozzles is 600 m/s. The turbines use 3500 kg/h of steam. The absolute velocity at exit is along the axis of turbine. Determine. i) The inlet and exit angles of the blades ii) The power output of the turbine iii) The diagram efficiency iv) The axial thrust Assume inlet and outlet angles to be equal.
13.	In a De-laval turbine steam issues from the nozzle with a velocity of 1200m/sec. the nozzle angle is 20° , the mean blade velocity is 400m/sec & the inlet & outlet angles of blades are equal. The mass of steam flowing through the turbine per hour is 900kg. Calculate i) Blade angles ii) Relative velocity of steam entering the blades iii) Tangential force on the blades & iv) Power developed, v) Efficiency. Assume that $K = 0.8$
14.	In a stage of an impulse turbine provided with a single row wheel, the mean diameter of blade ring is 0.08 m and the speed of rotation is 3000 rpm. The steam issues from the nozzle with a velocity of 300 m/s and the nozzle angle is 20°. The blades are equiangular and blade coefficient is 0.86. Determine the power developed in the blades when the axial thrust on the blade is 140 N.
	A single stage of simple impulse turbine produces 120 kW at blade speed of 150 m/s when steam mass flow rate is 3 kg/s. Steam enters moving blade at 350 m/s and leaves the stage axially.

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	Considering velocity coefficient of 0.9 and smooth steam entry without shock into blades, determine the nozzle angle and blade angles. Solve using velocity diagram
15.	Prove that $\eta_{b,max} = \frac{2 \cos^2 \alpha_1}{1 + \cos \alpha_1}$
16.	Define steam condenser and classification of steam condenser
17.	Differentiate between the Jet and surface steam condenser
18.	Explain parallel flow type jet ,counter flow jet condenser , high level jet condenser and ejector condenser
19.	Explain with neat sketch following surface condensers (i)Down Flow (ii) Central Flow (iii) Regenerative (iv) Evaporative
20.	Define following terms (1) Vacuum efficiency (2) Condenser efficiency (2) Air pump in condenser
21.	Explain the working of jet condenser with neat sketch
22.	Explain in brief necessity of cooling towers in steam power plant
23.	Explain with neat sketch induced draught cooling tower.
24.	Define cooling tower and explain types of cooling tower
Unit No.5	
1.	Define air compressor and How compressor are classified?
2.	Described the working of a single stage reciprocating air compressor
3.	Explain the working of centrifugal compressors and Axial flow compressor with neat sketch
4.	Explain the construction and working of a sliding vane air compressor.
5.	Derive an expression for the work done by single stage single acting reciprocating air compressor.

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6.	How actual indicator diagram for a single stage compressor differs from theoretical diagram
7.	Drive an expression for the work done by single stage single acting reciprocating air compressor with Clearance Volume
8.	What is multistage air compressor? Explain with neat sketch two stage air compressor with intercooler
9.	Define following terms 1) Slip factor 2) Power input factor 3) Free air delivery (FAD) 4) Volumetric efficiency 5) Mechanical efficiency 6) Isothermal efficiency 7) Mean effective pressure
10.	What is the difference between rotary and reciprocating air compressor
11.	Explain velocity diagram of centrifugal compressor
12.	A single stage reciprocating air compressor takes in $8\text{ m}^3/\text{min}$ of air at 1 bar and 30°C and delivers it at 6 bar. The clearance is 5% of the stroke. The expansion and compression are polytropic with the value of $n=1.3$. Calculate: (a) the temperature of delivered air; (b) volumetric efficiency, and (c) Power of the compressor.
13.	A single stage double acting air compressor delivers 5 m^3 of free air per minute at 1 ata and 20°C to 7.5 bar with the following data; R.P.M. = 300; Mechanical efficiency = 0.9; Pressure loss in passing through intake valves = 0.04 bar; Temperature rise of air during suction stroke = 12°C ; Clearance volume = 5% of stroke volume; Index of compression and expansion, $n=1.3$; Length of the stroke = 1.3 times the cylinder diameter. Calculate: 1. Power input to the shaft; 2. The volumetric efficiency; and 3. The cylinder diameter.
14.	Two stages, single acting air compressor compresses air to 20bar. The air enters the L.P cylinder at 1bar and 27°C and leaves it at 4.7bar. The air enters the H.P. cylinder at 4.5bar and 27°C . the size of the L.P cylinder is 400mm diameter and 500mm stroke. The clearance volume In both cylinder is 4% of the respective stroke volume. The compressor runs at 200rpm, taking index of compression and expansion in the two cylinders as 1.3, estimate 1. The indicated power required to run the compressor; and 2. The heat rejected in the intercooler per minute